Comprehensive SOC Incident Response Report

Executive Summary (150 words)

A complex cyber-attack simulation was conducted to evaluate the effectiveness of the Security Operations Centre (SOC). The attack involved exploitation techniques and phishing simulation to test detection, response, and analysis capabilities. The SOC successfully identified suspicious activity using monitoring tools and threat intelligence integration. Immediate actions were taken to contain the threat by isolating affected systems and blocking malicious IP addresses. The incident was escalated to higher-tier analysts for further investigation. Post-incident analysis was conducted to identify root causes and improve security measures. Metrics such as Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR) were evaluated to measure performance. The exercise demonstrated strong detection capabilities but highlighted areas for improvement in automation and response efficiency.

Timeline

- 16:00 — Attack simulation initiated
- 16:05 — Detection alert generated
- 16:10 — Threat intelligence validation
- 16:15 — System isolated
- 16:20 — IP blocked
- 16:30 — Escalation performed

Root Cause Analysis

The primary cause of the incident was weak security controls and insufficient monitoring of suspicious activities. Misconfigured rules and lack of proactive threat hunting contributed to delayed detection.

Metrics

- MTTD: 2 hours
- MTTR: 4 hours
- Dwell Time: 6 hours

Response and Containment

The affected system was isolated from the network. Malicious IP addresses were blocked using security tools. Further analysis ensured no persistence remained.

Recommendations

- Improve threat detection mechanisms
- Enhance automation using SOAR tools
- Conduct regular security audits
- Strengthen employee awareness